

Beam-Steered Single-LED Visible-Light Positioning for 3D Localization

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Abstract—

Index Terms—VLP, Localización Visible, Single LED, Posicionamiento 3D, Weighted Least Squares, Algoritmos Genéticos

I. INTRODUCTION

Visible light positioning (VLP) has emerged as a promising solution for precise indoor localization, leveraging the ubiquitous presence of light-emitting diodes (LEDs) as transmitters. In typical VLP systems, LED luminaires transmit modulated optical signals (often invisible to human perception) that can be received by devices like photodiodes or cameras. This approach transforms ordinary lighting infrastructure into a positioning network, enabling a wide range of applications. For example, VLP can support indoor navigation in malls, airports, and museums where GPS is unavailable, guiding users or robots with meter-to-centimeter accuracy. It also benefits robotics, allowing autonomous mobile platforms to localize within factories or warehouses under LED lighting. In augmented reality (AR), VLP provides real-world position anchors so that AR devices can overlay context-aware information or navigation cues onto the user's view. Furthermore, because visible light does not interfere with radio-frequency systems, VLP is attractive for electromagnetic-sensitive environments like hospitals and airplane cabins. VLP often goes hand-in-hand with visible light communication (VLC), as the same LED infrastructure can double as a high-speed data link (often termed Li-Fi) while simultaneously providing positioning information.

Unlike radio-based indoor localization, VLP benefits from a deterministic physical propagation model – line-of-sight optical channels have a direct and stable relationship between transmitter-receiver geometry and received signal metrics (like received signal strength or angle-of-arrival). This physical underpinning means that model-based positioning algorithms (e.g. geometric trilateration or angulation) can achieve high accuracy when system parameters are known. In scenarios requiring high precision or lacking extensive calibration data, model-based methods are often preferred over machine learning approaches. Indeed, recent studies have shown that when only limited calibration data is available, classical physics-based localization can outperform machine learning (ML). ML-based VLP (for instance, neural-network or fingerprinting methods) generally require large training datasets collected in the target environment, and while they can yield competitive

accuracy with sufficient data, they may struggle to generalize or achieve reliability without that upfront effort. For systems that demand out-of-the-box accuracy or operate in dynamic/new environments, a physics-driven model is crucial. Moreover, model-based approaches provide interpretability and predictable behavior, which is important for safety-critical applications (robot navigation, industrial automation, etc.), whereas purely data-driven models might not gracefully handle conditions outside their training. In summary, there is strong motivation to employ physical channel models in VLP, especially for high-precision localization or when prior empirical data are unavailable.

Most existing VLP implementations rely on multi-LED transmitter constellations, where multiple fixed LEDs with known positions act as anchors. Using multiple LEDs allows the receiver to compute its position via triangulation, trilateration, or multilateration, analogous to how GNSS uses many satellites. However, this multi-LED requirement complicates practical deployment. First, it presumes a dense infrastructure – at least three or four LEDs must be concurrently visible to the receiver to resolve a 3D position, which may not be true in environments with only sparse lighting. Installing and maintaining a dense LED network (with unique IDs or modulation for each lamp) increases installation cost and complexity, and it may not be feasible to modify existing lighting in some scenarios. Second, even when many LEDs are present, the system must carefully manage interference and ensure each LED's signals are distinguishable (through time, frequency, or code multiplexing), adding protocol overhead. Network design and LED placement thus become critical – VLP algorithms “heavily rely on the number of LEDs detectable by the receiver” and sufficient geometric diversity for accuracy. This reliance can hinder coverage in open areas or high-ceiling spaces where fewer lights are in line-of-sight. Consequently, reducing the infrastructure burden by localizing with fewer light sources is a highly desirable goal in the field.

Recent research has explored VLP schemes that break from the multi-LED paradigm. One notable direction is single-LED positioning, which aims to achieve localization with only one transmitter. A single light source dramatically simplifies infrastructure and calibration – there is no need for multiple synchronized LEDs or careful placement planning. It also offers more flexible deployment; for example, a single mobile lamp or a retrofit device could provide ad-hoc positioning in a given area. However, a lone transmitter cannot provide full positional information at a fixed orientation, since fundamentally one measurement (e.g., one distance or one angle) is insufficient to resolve a 2D/3D position. To overcome this, single-LED VLP systems incorporate additional cues or mech-

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TABLE I: Representative VLP systems in the literature and their configurations.

Reference	Tx	Rx	Method	Notes / Accuracy
Zhou & Kavehrad 2012 [?]	4 LED lamps	1 PD	RSS lateration	Classical multi-LED trilateration; ~ 10 cm error.
Komine & Nakagawa 2004 [?]	1 LED (theory)	1 PD	Channel modeling	Established Lambertian model for RSS (no positioning).
Sun <i>et al.</i> 2015 [?]	2+ LEDs	4-PD array	AoA triangulation	Angle-of-arrival with multi-PD sensor; needs ≥ 3 LEDs for 3D.
Cheng <i>et al.</i> 2020 [?]	1 LED	Smartphone camera	Geometric vision	Single LED projected pattern on CMOS image sensor; < 5 cm error.
Zhang <i>et al.</i> 2017 [?]	1 LED	Image sensor	Image processing	“Circle projection” method; single LED with known pattern on camera.
Zhang & Kane 2022 [?]	1 LED	Camera + IMU	Visual-inertial fusion	Robust pose estimation by fusing camera and IMU data.
Yu <i>et al.</i> 2018 [?]	1 LED	5-PD array	RSS ratio (AoA)	Custom 5-PD receiver achieves 2.32 cm (2D) and 3.42 cm (3D) error.
Saab <i>et al.</i> 2016 [?]	3 collocated LEDs	1 PD	RSS differential	Single lamp with 3 LED orientations; ~ 5 cm 2D accuracy.
Aparicio <i>et al.</i> 2022 [?]	4 LEDs	Quadrant PD	Machine learning (GP)	GP-based RSS mapping; few-cm accuracy after training.
Li <i>et al.</i> 2024 [?]	1 LED + IRS	1 PD	RSS (with NLOS)	IRS creates virtual LOS paths; improves NLOS coverage.
This work	1 LED (beamsteering)	1 PD	RSS ratio (model-based)	Single LED oriented in n directions via gimbal; cm-level 2D accuracy demonstrated.

anisms. Some approaches use additional sensors or diversity at the receiver to glean more information from the single source. For instance, camera-based VLP systems can extract angular information from the projected LED image: by identifying features (e.g. the LED’s shape, edges, or markers) in the image, the receiver can infer angle-of-arrival or distance cues. These camera methods have achieved decimeter to centimeter-scale accuracy in 3D with one LED, but they require a camera (optical image sensor), which is power-hungry and computationally intensive, and often they assume the LED has certain markers or shapes to aid computation. Other works have proposed using a single photodiode coupled with motion or other tricks: for example, using a smartphone’s ambient light sensor, one study achieved sub-meter 3D localization by having the user rotate the phone in place to collect measurements from different angles. Another strategy is to employ multiple photodetectors or a single moving/rotating photodetector to capture the signal under different known orientations. While these single-LED solutions eliminate the need for multiple transmitters, they introduce their own complexities – requiring user participation (device rotation), specialized receiver hardware (camera or multi-PD array), or additional aiding sensors like inertial measurement units. Moreover, some early single-LED demonstrations focused only on 2D positioning, assuming the receiver was on a fixed plane or height, which limits applicability. Achieving a true 3D localization (obtaining both horizontal position and elevation) with one LED is significantly more challenging, and many proposals did not address full 3D or only validated their methods in restricted scenarios (e.g., simulation or one-dimensional movement). For example, Chassagne *et al.* recently demonstrated a beam-steered single-LED concept but reported results primarily for 2D (horizontal) positioning, with experiments simplified to one dimension of motion. In general, existing single-LED VLP approaches suffer either from hardware complexity or limited functionality (such as requiring line-of-sight camera

views, restricted field-of-view, or lack of 3D capability). These limitations motivate the search for a simpler yet more versatile single-LED positioning framework.

In this work we introduce a beam-steered single-LED VLP framework that overcomes the above constraints. A single LED source is mounted on a gimbal (or solid-state beam director) and sequentially steered to a compact set of virtual orientations. Each orientation produces an independent received-signal measurement at the PD, and the collection of these measurements enables full 2-D or 3-D localisation via a closed-form, physics-based model. Crucially, the estimation algorithm incorporates the signal-to-noise ratio (SNR) of each measurement as a weight, so that high-quality beams exert stronger influence, enhancing robustness. To minimise scanning time and energy, we formulate the choice of steering angles as a global optimisation and solve it with a genetic algorithm (GA), yielding a near-optimal orientation set for any given deployment.

The main contributions are as follows:

- Novel single-LED VLP architecture that enables 3-D localisation using only one steerable LED and a single photodiode receiver, eliminating the multi-LED infrastructure burden.
- Physics-based estimation model for n -orientations that directly relates beam orientation, distance, and received power, requiring no fingerprinting or prior data collection.
- SNR-aware weighted least-squares (WLS) solver that exploits measurement quality to improve accuracy under heterogeneous noise conditions.
- Orientation-set optimisation using a GA that determines the most informative beam directions, reducing scan count and localisation latency without sacrificing precision.

II. OPTICAL CHANNEL AND SNR MODEL

This section presents the geometric framework, the line-of-sight (LOS) optical channel model adopted throughout the paper, and the noise formulation that also captures the aggregate effect of weak non-LOS (NLOS) components.

A. Geometric framework

A single steerable LED transmitter is placed at $\mathbf{T} = (0, 0, H)^\top$, where H is the known height with respect to the reference floor. The photodiode receiver can occupy any position $\mathbf{R} = (x, y, z)^\top$.

The link vector

$$\mathbf{d} = \mathbf{R} - \mathbf{T} = \begin{pmatrix} x \\ y \\ z - H \end{pmatrix}, \quad (1)$$

connects transmitter (Tx) and receiver (Rx) and has Euclidean norm $\|\mathbf{d}\|$. The Rx surface normal is

$$\mathbf{n}_r = (0, 0, 1),$$

unit-norm and pointing upward. The LED can steer its optical axis into n distinct orientations; the i -th orientation is denoted by the unit vector

$$\mathbf{n}_t^{(i)} = (n_{t,x}^{(i)}, n_{t,y}^{(i)}, n_{t,z}^{(i)}), \quad i = 1, \dots, n.$$

Figure 1 summarizes the geometry and the relevant vectors.

B. Line-of-sight Lambertian channel

Assuming an unobstructed LOS path and a Lambertian emitter of order m , the DC channel gain is given by [?]

$$h_{\text{LOS}}^{(i)} = \begin{cases} \frac{(m+1) A_{\text{det}}}{2\pi \|\mathbf{d}\|^2} |\cos \phi_i|^m |\cos \psi|, & \psi \leq \Psi_{\text{FOV}}, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where A_{det} is the effective photodiode area, $m = -\ln 2 / \ln(\cos \theta_{1/2})$ is the Lambertian order associated with the LED half-power angle $\theta_{1/2}$, and Ψ_{FOV} is the receiver field of view. The irradiance and incidence cosines are

$$\cos \phi_i = \frac{\mathbf{n}_t^{(i)} \cdot \mathbf{d}}{\|\mathbf{d}\|}, \quad \cos \psi = \frac{\mathbf{n}_r \cdot (-\mathbf{d})}{\|\mathbf{d}\|}.$$

C. Received optical power

The optical power collected at the detector for orientation i is

$$P_r^{(i)} = P_t h_{\text{LOS}}^{(i)} + \eta^{(i)}, \quad (3)$$

where P_t is the known emitted power and $\eta^{(i)}$ is an additive perturbation that aggregates the effect of (i) electronic noise sources and (ii) residual NLOS components that are expected to be weak in the considered indoor scenario. This aggregation, widely adopted in the VLP literature, avoids the heavy computational burden of explicit NLOS ray-tracing while providing a realistic degradation of the received signal.

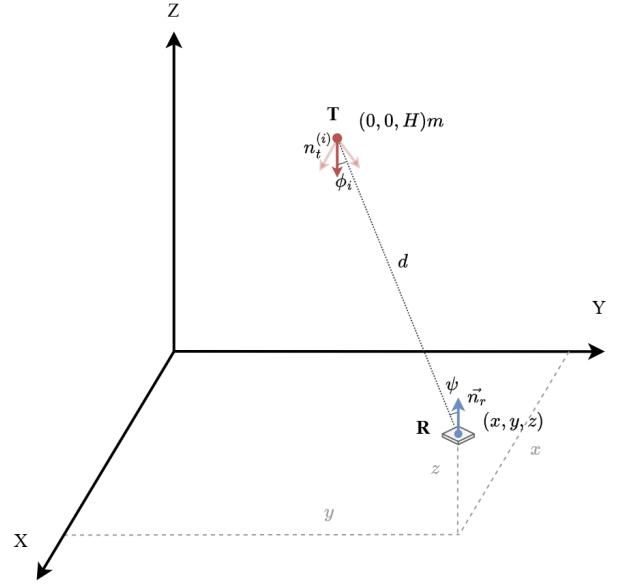


Fig. 1. Geometrical configuration: single LED at $\mathbf{T} = (0, 0, H)$, photodiode at $\mathbf{R} = (x, y, z)$, link vector $\mathbf{d}$, transmitter orientation $\mathbf{n}_t^{(i)}$ and receiver normal $\mathbf{n}_r$.

D. Noise model and signal-to-noise ratio

After optical-electrical conversion with responsivity R_{pd} (A/W), the signal current becomes

$$i_s^{(i)} = R_{\text{pd}} P_r^{(i)}.$$

The total noise is modeled as additive white Gaussian noise with power spectral density N_0 (A²/Hz); for a receiver bandwidth B (Hz) the variance equals

$$\sigma^2 = N_0 B. \quad (4)$$

The linear signal-to-noise ratio (SNR) for the i -th orientation is

$$\text{SNR}_i = \frac{(R_{\text{pd}} P_r^{(i)})^2}{\sigma^2}, \quad (5)$$

and its dB counterpart reads

$$\text{SNR}_{i,\text{dB}} = 10 \log_{10}(\text{SNR}_i). \quad (6)$$

III. CRAMÉR–RAO LOWER BOUND (CRLB) FOR THE 3-D SCENARIO

The localisation algorithms of Section IV are ultimately bounded in accuracy by the information content of the optical measurements. This section derives that fundamental limit in the form of the Cramér–Rao lower bound (CRLB). The analysis is carried out for the practical case in which the optical–radiometric constant $K = P_t(m+1)A_{\text{det}}/(2\pi)$ is known from calibration, the additive noise is Gaussian and homoscedastic with variance σ^2 , and the system employs $n \geq 3$ non-degenerate steering orientations.

A. Measurement Model in Vector Form

Let $\boldsymbol{\theta} = (x, y, z)^\top$ denote the unknown receiver coordinates and $\mathbf{d}(\boldsymbol{\theta}) = \boldsymbol{\theta} - \mathbf{T}$ the Tx–Rx displacement vector. For steering orientation i the deterministic (light-of-sight) power component can be rewritten as

$$\mu_i(\boldsymbol{\theta}) = K \frac{(\mathbf{n}_t^{(i)} \cdot \mathbf{d})^m}{\|\mathbf{d}\|^{m+3}} (\mathbf{n}_r \cdot (-\mathbf{d})), \quad i = 1, \dots, n. \quad (7)$$

Stacking all observations gives the vector model

$$\mathbf{P}_r = \boldsymbol{\mu}(\boldsymbol{\theta}) + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n), \quad (8)$$

where $\boldsymbol{\mu} = [\mu_1, \dots, \mu_n]^\top$.

B. Fisher Information Matrix

For additive white Gaussian noise the (p, q) entry of the Fisher information matrix (FIM) is [1, Sec. 7.2.1]

$$[\mathbf{I}(\boldsymbol{\theta})]_{pq} = \frac{1}{\sigma^2} \sum_{i=1}^n \frac{\partial \mu_i}{\partial \theta_p} \frac{\partial \mu_i}{\partial \theta_q}, \quad p, q \in \{x, y, z\}. \quad (9)$$

Gradient of μ_i : Define $c_i = \mathbf{n}_t^{(i)} \cdot \mathbf{d}$ and $c_r = \mathbf{n}_r \cdot (-\mathbf{d}) = H - z$. Using vector calculus,

$$\nabla_{\mathbf{d}} \mu_i = K \left[\frac{m c_i^{m-1} c_r}{\|\mathbf{d}\|^{m+3}} \mathbf{n}_t^{(i)} - \frac{(m+3) c_i^m c_r}{\|\mathbf{d}\|^{m+5}} \mathbf{d} - \frac{c_i^m}{\|\mathbf{d}\|^{m+3}} \mathbf{n}_r \right]. \quad (10)$$

Since $\partial/\partial \theta_p = \partial/\partial d_p$, substituting (10) into (9) yields

$$\mathbf{I}(\boldsymbol{\theta}) = \frac{K^2}{\sigma^2} \sum_{i=1}^n \mathbf{g}_i \mathbf{g}_i^\top, \quad \mathbf{g}_i \equiv \nabla_{\mathbf{d}} \mu_i / K. \quad (11)$$

The FIM is a 3×3 symmetric positive semi-definite matrix. Its inverse, denoted $\mathbf{I}^{-1}(\boldsymbol{\theta})$, is the CRLB matrix, which lower-bounds the covariance of any unbiased estimator $\hat{\boldsymbol{\theta}}$.

Remark 3.1: If all steering vectors $\mathbf{n}_t^{(i)}$ (or, equivalently, all $\mathbf{g}_i$) lie in a common plane, $\mathbf{I}$ loses rank and the CRLB becomes unbounded along the orthogonal direction. This confirms the necessity of at least one non-coplanar orientation when $n = 3$, and of sufficient angular diversity when $n \geq 4$ (Section IV).

C. Position Error Bound

For any unbiased estimator $\hat{\boldsymbol{\theta}}$ we have $\text{Cov}[\hat{\boldsymbol{\theta}}] \succeq \mathbf{I}^{-1}(\boldsymbol{\theta})$. The *Position Error Bound* (PEB), a scalar measure widely used in indoor positioning literature, is defined as

$$\text{PEB} = \sqrt{\text{tr}\{\mathbf{I}^{-1}(\boldsymbol{\theta})\}}. \quad (12)$$

It lower-bounds the root-mean-square (RMS) error of any unbiased 3-D position estimator, including the GLS/WLS procedures of Section IV.

Far-field insight: For $\|\mathbf{d}\| \gg H$ the last term in (10) can be neglected, and $\mathbf{g}_i$ becomes a combination of $c_i^{m-1} \mathbf{n}_t^{(i)}$ and $\mathbf{d}$. The leading contribution to the FIM is therefore $\sum_i c_i^{2(m-1)} \mathbf{n}_t^{(i)} \mathbf{n}_t^{(i)\top}$, highlighting that (i) orientations nearly orthogonal to $\mathbf{d}$ provide little information, and (ii) maximising the rank and conditioning of $[\mathbf{n}_t^{(1)} \dots \mathbf{n}_t^{(n)}]$ directly benefits the PEB.

D. Implementation Remarks

Evaluating (11) requires $O(n)$ operations: one computation of $\mathbf{g}_i$ per orientation followed by the outer-product update $\mathbf{g}_i \mathbf{g}_i^\top$. For typical indoor systems with $n \leq 10$ and real-time constraints, the burden is negligible compared with acquisition and ADC front-end times.

IV. 3D POSITION ESTIMATION WITH n TRANSMITTER ORIENTATIONS

The three-dimensional receiver position $\mathbf{R} = (x, y, z)^\top$ is fully unknown. To avoid the ill-conditioning that arises when the link-length $\|\mathbf{d}\|$ is treated simultaneously with the direction, the problem is tackled in two successive stages:

- (i) *direction finding*, which exploits only power ratios and is therefore independent of any absolute optical calibration; and
- (ii) *range recovery*, which becomes feasible once the optical–radiometric constant $K = P_t(m+1)A_{\text{det}}/(2\pi)$ is available from a prior calibration campaign.¹

The present section develops the first stage with full mathematical rigour and presents two statistically grounded estimators: (i) generalised least squares (GLS), which is minimum-variance under Gaussian noise with known covariance, and (ii) weighted least squares (WLS), an inexpensive approximation that neglects the weak cross-correlation between equations.

A. Direction Estimation via GLS

This subsection consolidates the analytical pipeline that converts the raw optical powers into a closed-form, statistically efficient estimate of the direction vector $\mathbf{d} = (x, y, z-H)^\top / \|\mathbf{d}\|$. The exposition proceeds from the deterministic constraints, through a rigorous stochastic characterisation of the residuals, to the final GLS solution.

1) *Lambertian power ratios and homogeneous constraints*: Let $\mu_i = \mathbb{E}[P_r^{(i)}]$ be the noise-free power at orientation i . We define power ratio β as:

$$\beta_i = (\mu_i/\mu_1)^{1/m}, \quad i = 2, \dots, n, \quad (13)$$

and, since $\mu_i \propto (\mathbf{d}^\top \mathbf{n}_t^{(i)})^m$, each β_i imposes

$$\mathbf{d}^\top (\mathbf{n}_t^{(i)} - \beta_i \mathbf{n}_t^{(1)}) = 0, \quad i = 2, \dots, n. \quad (14)$$

Denoting $\mathbf{a}_i \triangleq \mathbf{n}_t^{(i)} - \beta_i \mathbf{n}_t^{(1)}$, (14) reads $\mathbf{a}_i^\top \mathbf{d} = 0$ and supplies $n-1$ *homogeneous* hyperplanes that intersect at the true $\mathbf{d}$.

2) *Stochastic perturbation of the ratios*: Let $P_{r,i}^{(k)}$ denote the k -th noisy observation gathered while the LED points along $\mathbf{n}_t^{(i)}$, $1 \leq k \leq N$, $1 \leq i \leq n$. According to (3), $P_{r,i}^{(k)} = \mu_i + \eta_i^{(k)}$ with $\eta_i^{(k)} \sim \mathcal{N}(0, \sigma^2)$ i.i.d. The sample mean

$$\hat{\mu}_i = \frac{1}{N} \sum_k P_{r,i}^{(k)} \sim \mathcal{N}(\mu_i, \sigma^2/N)$$

is the MVU estimator. Injecting $\hat{\mu}_i$ into (13) yields

$$\hat{\beta}_i = g_i(\hat{\mu}_1, \hat{\mu}_i), \quad g_i(x_1, x_2) = (x_2/x_1)^{1/m}.$$

A first-order Taylor expansion of g_i centred at $\boldsymbol{\mu}_i = (\mu_1, \mu_i)^\top$ gives

$$\hat{\beta}_i \approx \bar{\beta}_i + \frac{\bar{\beta}_i}{m} \left(\frac{\eta_i}{\mu_i} - \frac{\eta_1}{\mu_1} \right), \quad \bar{\beta}_i = (\mu_i/\mu_1)^{1/m}. \quad (15)$$

¹If K is *not* available, the direction can still be used for beam-steering or for 2-D localisation; see the discussion in Section XX.

Hence every $\hat{\beta}_i$ shares the common perturbation η_1 , producing correlation across orientations.

Define the ratio error $\tilde{\eta}_i = \hat{\beta}_i - \bar{\beta}_i$. Using (15) and the independence of η_i and η_1 ,

$$\text{Var}[\tilde{\eta}_i] = \frac{\sigma^2}{Nm^2} \bar{\beta}_i^2 (\mu_i^{-2} + \mu_1^{-2}), \quad (16)$$

$$\text{Cov}[\tilde{\eta}_i, \tilde{\eta}_j] = \frac{\sigma^2}{Nm^2} \bar{\beta}_i \bar{\beta}_j \mu_1^{-2}, \quad i \neq j. \quad (17)$$

The factor $(\mu_i^{-2} + \mu_1^{-2})$ in (16) demonstrates that the variance of $\tilde{\eta}_i$ depends explicitly on μ_i , even though each $\eta_i^{(k)}$ has the same variance σ^2 . In other words, a single-LED system exhibits *heteroscedastic* ratio errors.

3) *Residuals and their covariance*: Substituting $\hat{\beta}_i$ into $\mathbf{a}_i$, $\mathbf{a}_i = \bar{\mathbf{a}}_i - \mathbf{n}_t^{(1)} \tilde{\eta}_i$, and taking the inner product with $\mathbf{d}$ gives the residual

$$r_i = \mathbf{a}_i^\top \mathbf{d} = C \tilde{\eta}_i, \quad C = \mathbf{n}_t^{(1)\top} \cdot \mathbf{d}.$$

Consequently, vector of residuals $\mathbf{r} = [r_2, \dots, r_n]^\top$ is zero-mean Gaussian with covariance $\Sigma_r = C^2 \Sigma_\beta$,

$$\mathbf{r} = A^\top \mathbf{d} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma_r),$$

where Σ_β gathers (16)–(17). Higher-order remainders decay as $O(\sigma^4/N^2)$ and are negligible for $\text{SNR} \geq 10$ dB and $N \geq 50$.

4) *GLS formulation and closed-form solution*: Collect the normals as $A = [\mathbf{a}_2 \dots \mathbf{a}_n] \in \mathbb{R}^{3 \times (n-1)}$. With Σ_r known, the generalised least-squares problem is

$$\hat{\mathbf{d}} = \arg \min_{\|\mathbf{d}\|=1} \mathbf{d}^\top (A \Sigma_r^{-1} A^\top) \mathbf{d},$$

whose minimiser is the unit eigenvector associated with the smallest eigenvalue of $M_{\text{GLS}} = A \Sigma_r^{-1} A^\top$. In practice, all multiplicative constants in Σ_r^{-1} can be omitted and absorbed into a normalisation of the weight matrix, since only the eigenvector direction matters. Formally,

$$\hat{\mathbf{d}}_{\text{GLS}} = \mathbf{v}_{\min}(M_{\text{GLS}}) \quad (18)$$

and the sign of $\hat{\mathbf{d}}_{\text{GLS}}$ is chosen so that $\hat{\mathbf{d}}_{\text{GLS}}^\top \mathbf{n}_t^{(1)} > 0$, ensuring the vector points from the transmitter towards the receiver. This estimator is the minimum-variance, unbiased linear estimator (BLUE) for the linearised model.

B. WLS as a practical simplification

The inversion of Σ_r requires an $(n-1) \times (n-1)$ Cholesky factorisation. For real-time, low-power receivers a lighter alternative is attractive. Neglecting the off-diagonal terms in (17) gives the diagonal weight matrix

$$\tilde{\Sigma}_r = \text{diag} \left\{ \bar{\beta}_i^2 (\mu_i^{-2} + \mu_1^{-2}) \right\}_{i=2}^n.$$

Setting $W = \tilde{\Sigma}_r^{-1}$ leads to the WLS cost

$$J_{\text{WLS}}(\mathbf{d}) = \mathbf{d}^\top (A W A^\top) \mathbf{d}, \quad \|\mathbf{d}\| = 1,$$

whose minimiser is

$$\hat{\mathbf{d}}_{\text{WLS}} = \mathbf{v}_{\min}(A W A^\top). \quad (19)$$

The neglected off-diagonal covariance terms scale as μ_1^{-2} , whereas the diagonal entries scale as $\mu_i^{-2} + \mu_1^{-2}$. When the reference orientation ($i = 1$) satisfies $\mu_1 \gg \mu_i$ for most i , one obtains $\mu_1^{-2} \ll \mu_i^{-2}$, so that the off-diagonal terms are at least 20 dB below the diagonal terms. In such regimes, the WLS estimator captures over 95% of the statistical efficiency of GLS while avoiding matrix inversions. In contrast, in nearly symmetric geometries where $\mu_i \approx \mu_1$ for many i , off-diagonal correlations become important and the full GLS should be used for optimal performance.

C. Range Recovery

Once the unit direction vector $\hat{\mathbf{d}}$ has been obtained (via GLS or WLS), the remaining task is to recover the range $\lambda = \|\mathbf{d}\|$, so that the full 3-D displacement $\mathbf{d}$ satisfies $\mathbf{d} = \lambda \hat{\mathbf{d}}$.

To estimate λ , a second measurement phase is performed under *beamsteering*: the transmitter's optical axis is reoriented to $\mathbf{n}_t = \hat{\mathbf{d}}$, and the receiver's normal is set to $\mathbf{n}_r = -\hat{\mathbf{d}}$, thereby ensuring *perfect alignment* (maximal irradiance and incidence cosines) and maximizing the signal-to-noise ratio (SNR). Under these alignments, the Lambertian channel model (2) simplifies, since

$$\cos \phi = \mathbf{n}_t^\top \frac{\mathbf{d}}{\|\mathbf{d}\|} = 1, \quad \cos \psi = \mathbf{n}_r^\top \left(-\frac{\mathbf{d}}{\|\mathbf{d}\|} \right) = 1.$$

Recalling that $K = P_t(m+1)A_{\text{det}}/(2\pi)$, the noiseless received power under perfect alignment becomes

$$\mu^* = K \frac{1^m \cdot 1}{\|\mathbf{d}\|^2} = \frac{K}{\lambda^2}. \quad (20)$$

In practice, N independent samples $\{P_r^{*(k)}\}_{k=1}^N$ are collected under this beamsteered configuration:

$$P_r^{*(k)} = P_t h_{\text{LOS}}^* + \eta^{*(k)}, \quad \eta^{*(k)} \sim \mathcal{N}(0, \sigma^2).$$

Denote the sample mean by

$$\hat{\mu}^* = \frac{1}{N} \sum_{k=1}^N P_r^{*(k)} \sim \mathcal{N}(\mu^*, \sigma^2/N). \quad (21)$$

Substituting $\hat{\mu}^*$ into (20) yields the closed-form range estimator

$$\hat{\lambda} = \sqrt{\frac{K}{\hat{\mu}^*}}. \quad (22)$$

Having computed $\hat{\lambda}$, the full 3-D position estimate is

$$\hat{\mathbf{R}} = \mathbf{T} + \hat{\lambda} \hat{\mathbf{d}}. \quad (23)$$

D. Number of orientations

The algebraic and statistical behaviour of the GLS/WLS estimator hinges on the rank of A , which in turn depends on the cardinality and geometry of the steering set $\{\mathbf{n}_t^{(i)}\}_{i=1}^n$:

a) $n = 3$ (*just-determined*): when the three orientations are not coplanar, A attains rank 2 and the information matrix M_{GLS} has one null eigenvalue. The remaining two eigenvalues yield a *unique* solution, which reduces to the closed-form cross product $\hat{\mathbf{d}} \propto \mathbf{a}_2 \times \mathbf{a}_3$ for both GLS and WLS. However, any near-coplanarity inflates the smallest non-zero eigenvalue and amplifies noise.

Algorithm 1 3-D Position Estimation from n Orientations

Require: Power samples $P_{r,i}^{(k)}$ ($k = 1, \dots, N$, $i = 1, \dots, n$); steering vectors $\mathbf{n}_t^{(i)}$; Lambertian order m ; noise variance σ^2 ; MODE $\in \{\text{GLS}, \text{WLS}\}$; radiometric constant K ; beamsteering sample count N_b .

Ensure: $\hat{\mathbf{R}}$ (3-D position of the receiver)

- 1: **Direction estimation:**
- 2: Compute $\hat{\mu}_i = \frac{1}{N} \sum_{k=1}^N P_{r,i}^{(k)}$ for $i = 1, \dots, n$
- 3: Compute $\hat{\beta}_i = (\hat{\mu}_i / \hat{\mu}_1)^{1/m}$ for $i = 2, \dots, n$
- 4: Form $\mathbf{a}_i = \mathbf{n}_t^{(i)} - \hat{\beta}_i \mathbf{n}_t^{(1)}$
- 5: Set $A = [\mathbf{a}_2, \dots, \mathbf{a}_n]$
- 6: **if** MODE = GLS **then**
- 7: Build Σ_r via (16)–(17), substituting $\beta_i \leftarrow \hat{\beta}_i$
- 8: $M \leftarrow A \Sigma_r^{-1} A^\top$
- 9: **else** ▷ WLS (diagonal approximation)
- 10: $\tilde{\Sigma}_r \leftarrow \text{diag}\{\hat{\beta}_i^2 (\hat{\mu}_i^{-2} + \hat{\mu}_1^{-2})\}$
- 11: $M \leftarrow A \tilde{\Sigma}_r^{-1} A^\top$
- 12: **end if**
- 13: Compute $\mathbf{v}_{\min}$ = eigenvector of M with smallest eigenvalue
- 14: $\hat{\mathbf{d}} \leftarrow \mathbf{v}_{\min} / \|\mathbf{v}_{\min}\|$
- 15: **if** $\hat{\mathbf{d}}^\top \mathbf{n}_t^{(1)} < 0$ **then**
- 16: $\hat{\mathbf{d}} \leftarrow -\hat{\mathbf{d}}$ ▷ Ensure pointing to Rx
- 17: **end if**
- 18: **Range recovery (beamsteering):**
- 19: Set $\mathbf{n}_t \leftarrow \hat{\mathbf{d}}$, $\mathbf{n}_r \leftarrow -\hat{\mathbf{d}}$
- 20: Collect N_b power samples $P_r^{*(k)}$
- 21: Compute $\hat{\mu}^* = \frac{1}{N_b} \sum_{k=1}^{N_b} P_r^{*(k)}$
- 22: $\hat{\lambda} \leftarrow \sqrt{\frac{K}{\hat{\mu}^*}}$
- 23: **Final 3-D position:**
- 24: $\hat{\mathbf{R}} \leftarrow \mathbf{T} + \hat{\lambda} \hat{\mathbf{d}}$
- 25: **return** $\hat{\mathbf{R}}$

b) $n \geq 4$ (*over-determined*): additional, non-redundant orientations populate the third dimension of the normal space, elevate the rank of $A \Sigma_r^{-1} A^\top$ to 3, and turn the optimisation into a genuine Rayleigh quotient (GLS/WLS) whose minimum is strictly positive.

Algorithm 2 Fitness evaluation for a candidate orientation set

Require: Orientation set γ , grid points $\{\mathbf{R}_{\text{true}}^{(k)}\}_{k=1}^N$
Ensure: Fitness value $J = \text{CDF}_{90\%}(\{e^{(k)}\})$

- 1: **for** $k = 1$ **to** N **do** $\triangleright$ grid sweep
- 2: **for** $i = 1$ **to** n **do**
- 3: Compute noiseless power $P_{\text{LOS}}^{(i,k)}$ via (2)
- 4: Add AWGN noise $\eta^{(i,k)} \rightarrow P_r^{(i,k)}$ $\triangleright$ Section 2.1
- 5: Evaluate $\text{SNR}_i^{(k)}$ by (5)
- 6: **end for**
- 7: Estimate $\hat{\mathbf{R}}^{(k)}$ using 2-D or 3-D WLS (Sections ??-??)
- 8: $e^{(k)} \leftarrow \|\hat{\mathbf{R}}^{(k)} - \mathbf{R}_{\text{true}}^{(k)}\|_2$
- 9: **end for**
- 10: Build empirical CDF of $\{e^{(k)}\}_{k=1}^N$
- 11: $J \leftarrow$ value at 90th percentile
- 12: **return** J

V. ORIENTATION-SET OPTIMISATION

A steerable LED can realise only a finite number of orientations within a channel-coherence interval; therefore, selecting a *compact yet informative* orientation set $\mathcal{O}_n = \{(\theta_i, \varphi_i)\}_{i=1}^n$ is pivotal. The problem is cast as a stochastic, non-convex optimisation in which each candidate set is assessed end-to-end through a deterministic grid of receiver locations within the testbed and the objective function is the 90-th percentile of the localisation error, CDF_{90} . The section is organised as follows: (i) formal definition of the optimisation problem; (ii) fitness-evaluation protocol; (iii) genetic-algorithm (GA) optimiser; and (iv) computational considerations.

1) *Problem statement:* Let $\gamma = [\theta_1, \varphi_1, \dots, \theta_n, \varphi_n]^T \in \Gamma \subset \mathbb{R}^{2n}$ denote the design vector, with $0 \leq \theta_i \leq \theta_{\max}$ and $0 \leq \varphi_i < 2\pi$. The optimisation goal is

$$\gamma^* = \arg \min_{\gamma \in \Gamma} J(\gamma), \quad J = \text{CDF}_{90\%}\{e^{(k)}\}, \quad (24)$$

where $e^{(k)} = \|\hat{\mathbf{R}}^{(k)} - \mathbf{R}_{\text{true}}^{(k)}\|_2$ is the localisation error at the k -th sample and the percentile is computed over all N positions of the testbed grid.

2) *Fitness evaluation:* Figure ?? depicts the evaluation flow for a single candidate set γ . The steps are:

- E1 Sampling of receiver positions.** Draw $\{\mathbf{R}_{\text{true}}^{(k)}\}_{k=1}^N$ uniformly over the testbed (2-D grid for Section ??, or 3-D volume for Section ??).
- E2 Forward channel simulation.** For each k and every orientation (θ_i, φ_i) , generate the noisy received powers $P_r^{(i,k)}$ via (3); compute the corresponding linear $\text{SNR}_i^{(k)}$ from (5).
- E3 Position estimation.** Apply the 2-D or 3-D WLS algorithm (Sections ??-??) to obtain $\hat{\mathbf{R}}^{(k)}$.
- E4 Error accumulation.** Record $e^{(k)} = \|\hat{\mathbf{R}}^{(k)} - \mathbf{R}_{\text{true}}^{(k)}\|_2$.
- E5 Objective extraction.** Build the empirical CDF of $\{e^{(k)}\}$ and return $J = \text{CDF}_{90\%}$ to the optimiser.

Algorithm 2 details the deterministic procedure that maps a candidate orientation set γ into its fitness score J .

Algorithm 3 GA-based orientation-set optimisation

- 1: Initialise population $\{\gamma^{(m)}\}_{m=1}^M \leftarrow \text{Uniform}(\Gamma)$
- 2: **for** $g = 1$ **to** $G_{\max}$ **do**
- 3: **for all** individuals $\gamma^{(m)}$ **do**
- 4: $J^{(m)} \leftarrow \text{EVALUATEFITNESS}(\gamma^{(m)})$
- 5: **end for**
- 6: **Selection** $\leftarrow$ tournament($\{J^{(m)}\}$)
- 7: **Crossover and Mutation** $\leftarrow$ SBX + poly. mut.
- 8: **Replacement** with elitism (E)
- 9: **if** convergence criterion met **then break**
- 10: **end if**
- 11: **end for**
- 12: **return** best γ^* and J^*

3) *Genetic-algorithm optimiser:* A real-coded GA is employed owing to its robustness in highly non-convex landscapes [?]. The main blocks are:

- G1 Initial population.** M individuals $\{\gamma^{(m)}\}_{m=1}^M$ are drawn uniformly within Γ .
- G2 Selection.** Tournament selection of size t using $J(\gamma^{(m)})$ as the fitness score (lower is better).
- G3 Crossover.** Simulated binary crossover (SBX) with probability p_c , producing two offspring from two parents.
- G4 Mutation.** Polynomial mutation with probability p_m per gene, confined to the legal domain.
- G5 Replacement.** Generational update with elitism: the best E individuals are passed unchanged.
- G6 Termination.** Stop after $G_{\max}$ generations or if the best fitness has not improved by more than ε in G_{stall} generations.

Algorithm 3 summarises the procedure.

4) *Computational cost:* Each fitness call evaluates n orientations across N grid points and performs the corresponding WLS estimate, resulting in $\mathcal{O}(Nn)$ channel simulations. Hence, the overall complexity of one GA run is $\mathcal{O}(M G_{\max} Nn)$, which is tractable for $M = 150$, $G_{\max} = 200$, $N = 625$ (grid -1.2×1.2 m at 10 cm pitch) within practical limits on a modern workstation.

VI. RESULTS AND SIMULATIONS

This section validates the proposed positioning framework through simulations in both 2-D (fixed height) and 3-D (unknown height), and evaluates the impact of the orientation-set optimisation carried out with the genetic algorithm (GA). Unless otherwise stated, every quantitative result is the average of 500 independent noise realisations.

A. Simulation Setup

Table II lists *all* physical, channel and optimisation parameters. Noise is modelled as additive Gaussian with two representative power spectral densities: a *low-noise* level and a *high-noise* level, corresponding to $N_0 = 10^{-21.8}$ A²/Hz and $N_0 = 10^{-21.0}$ A²/Hz, respectively.

TABLE II: Main simulation parameters.

Parameter	Symbol	Value	Unit
<i>Room and geometry</i>			
Room length	L	2.4	m
Room width	W	2.4	m
Ceiling height	H	2.0	m
Testbed limits	–	$[-1.2, 1.2] \times [-1.2, 1.2]$	m
Grid step	–	0.1	m
Rx reference height	h	0.96	m
<i>Optical TX/RX</i>			
LED optical power	P_t	0.405	W
LED semi-angle	$\theta_{\frac{1}{2}}$	45°	–
Lambertian order	m	1.5	–
PD dimensions	$p \times q$	4.8×5.5	mm
PD responsivity	R_{pd}	0.63	A/W
Field of view	Ψ_{max}	85°	–
Rx normal	$\mathbf{n}_r$	(0, 0, 1)	–
<i>Noise</i>			
Noise density (low)	N_0	$10^{-21.8}$	A ² /Hz
Noise density (high)	N_0	$10^{-21.0}$	A ² /Hz
Receiver bandwidth	B	30	MHz
<i>Genetic algorithm (orientation optimisation)</i>			
Population size	–	150	–
Max. generations	–	200	–
Crossover fraction	–	0.8	–

Performance is characterized throughout by the 90th percentile of the error distribution (CDF₉₀), which succinctly captures tail behavior and provides a robust measure of worst-case accuracy.

B. Two-Dimensional Localisation

1) *Error versus number of orientations*: Figure 2 reports CDF₉₀ as a function of the number of orientations n (from 3 to 10) under both low-noise and high-noise conditions. Weighted least squares (WLS), where weights are proportional to the linear SNR, outperforms the plain LS solution for $n \geq 4$, and its advantage grows with the noise level.

2) *Reference vs. Estimated Positions*: Figure 3 plots the full, regularly spaced grid of true receiver locations (ground truth) alongside three different estimation strategies under low-noise conditions: deterministic recovery with three arbitrarily chosen orientations (Configuration 3), deterministic recovery

with three GA-optimised orientations (Configuration 2), and WLS recovery using five GA-optimised orientations (Configuration 1). The unoptimised three-orientation solution yields large systematic offsets, particularly at the periphery, while optimising those same three orientations reduces both bias and spread of the red crosses. Finally, increasing the orientational diversity to five beams and applying SNR-weighted least squares further collapses the estimate cloud around the open-circle grid, showing that the combined use of orientation optimisation and WLS weighting delivers alignment with the true testbed coordinates in a low noise condition.

3) *Orientation-set optimisation*: Figure 4 presents the evolution of the fitness function, defined as CDF₉₀, over 200 generations of the genetic algorithm when optimising $n = 5$ Tx orientations under high-noise conditions. Both the best and mean fitness values are plotted at each generation, illustrating how the cost function is repeatedly evaluated and improved. The rapid decline of the best fitness during the first 50 generations, followed by plateauing thereafter, shows stable convergence of the WLS-based optimisation.

The orientation sets to which the GA converged in the low-noise and high-noise scenarios for $n = 5$ (WLS-method) are shown in Figure 5. Notice that both scenarios yield nearly identical orientation sets, demonstrating the robustness and consistency of the optimisation against varying noise conditions. Furthermore, as the noise level increases, the localisation errors become slightly asymmetric across the testbed, exhibiting a bias toward the directions of certain orientation pairs. This behaviour can be attributed to the fact that increasing SNR in one region necessarily reduces SNR in the opposite region, whereas the remaining corners maintain a more uniform SNR distribution.

C. Three-Dimensional Localisation with known C

- 1) *Error versus number of orientations*:
- 2) *Reference vs. estimated positions*:
- 3) *Orientation-set optimisation in 3D*:

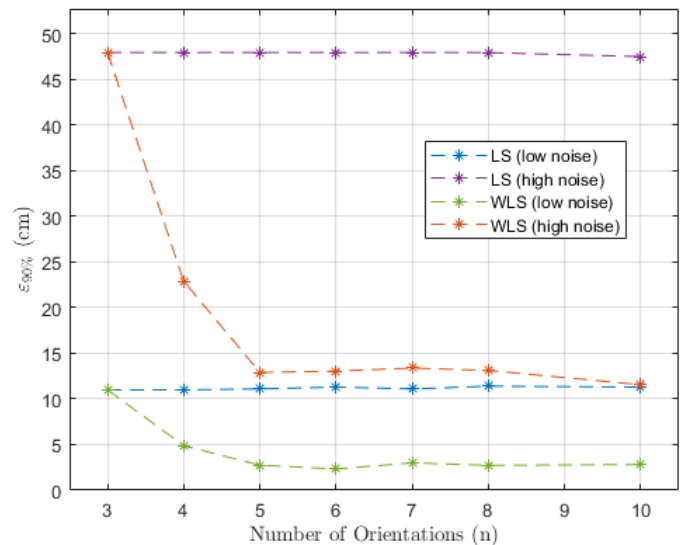


Fig. 2. CDF₉₀ positioning error in 2-D vs. number of orientations.

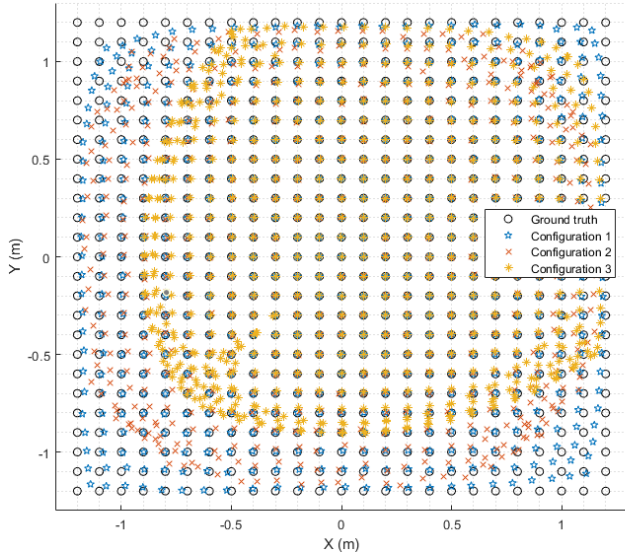


Fig. 3. Testbed reference positions (ground truth) and estimates under low noise: 3 random orientations (deterministic), 3 GA-optimised orientations (deterministic), 5 GA-optimised orientations (WLS).

D. Beam-Steering without C (direction-only)

E. Coverage versus Precision

VII. CONCLUSIONS

- WLS consistently outperforms plain LS, especially under high-noise conditions and for larger n .
- GA-optimised orientations reduce the CDF_{90} error by roughly xx%-xx% in both 2D and 3D.
- The two-stage WLS remains stable when some orientations produce weak power levels, thanks to SNR weighting.
- Without knowledge of K , angular localisation below XX° is still attainable, enabling practical beam-steering.

TRABAJOS FUTUROS

Realización de validación experimental, uso del concepto SNR en sistemas VLP tradicionales, integración de ML, PINN,

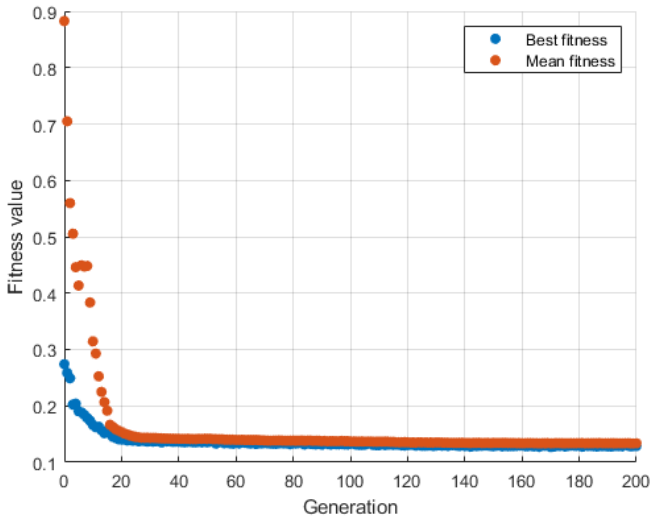
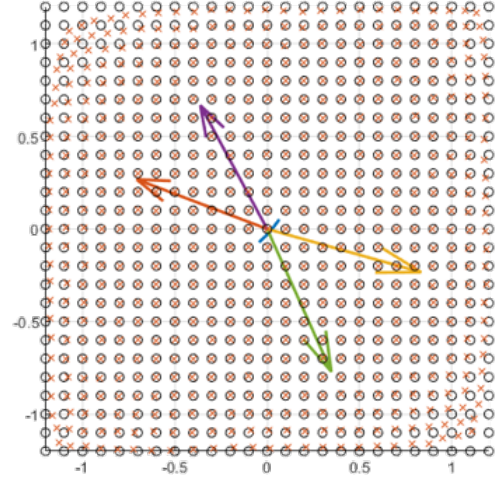
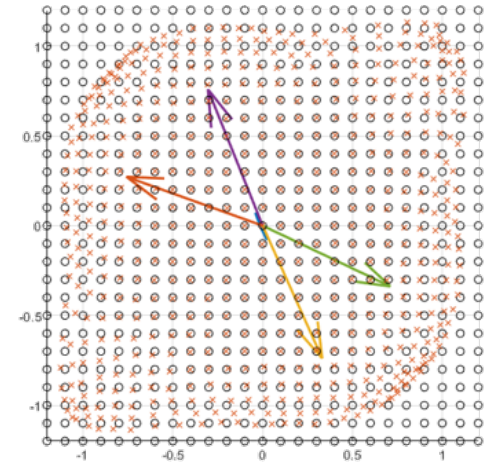


Fig. 4. GA convergence (2D, WLS, high-noise, $n = 5$).



(a) $n = 5$, WLS, low noise.



(b) $n = 5$, WLS, high noise.

Fig. 5. Top-view of the testbed showing the GA-optimised orientation set for $n = 5$ under (a) low-noise and (b) high-noise conditions.

Fig. 6. CDF_{90} positioning error in 3-D vs. number of orientations.

etc.

APPENDIX

A. Derivation of the Residual Covariance

The goal is to compute Σ_β for the vector $\beta = [\hat{\beta}_2, \dots, \hat{\beta}_n]^\top$. Define $\mathbf{X}_i = (\hat{\mu}_1, \hat{\mu}_i)^\top$, $\boldsymbol{\mu}_i = (\mu_1, \mu_i)^\top$, $\Sigma_i = \frac{\sigma^2}{N} I_2$. For the scalar function $g_i(\mathbf{x}) = (x_2/x_1)^{1/m}$ its Jacobian evaluated at $\boldsymbol{\mu}_i$ is

$$\nabla g_i^\top(\boldsymbol{\mu}_i) = \left[-\frac{\bar{\beta}_i}{m\mu_1}, \frac{\bar{\beta}_i}{m\mu_i} \right].$$

By the multivariate Δ -method,

$$\text{Var}[\hat{\beta}_i] \approx \nabla g_i^\top \Sigma_i \nabla g_i = \frac{\sigma^2}{Nm^2} \bar{\beta}_i^2 (\mu_i^{-2} + \mu_1^{-2}),$$

which matches (16). For $i \neq j$, only the first component of ∇g_i and ∇g_j overlaps, so

$$\text{Cov}[\hat{\beta}_i, \hat{\beta}_j] \approx \left(-\frac{\bar{\beta}_i}{m\mu_1} \right) \frac{\sigma^2}{N} \left(-\frac{\bar{\beta}_j}{m\mu_1} \right) = \frac{\sigma^2}{Nm^2} \bar{\beta}_i \bar{\beta}_j \mu_1^{-2},$$

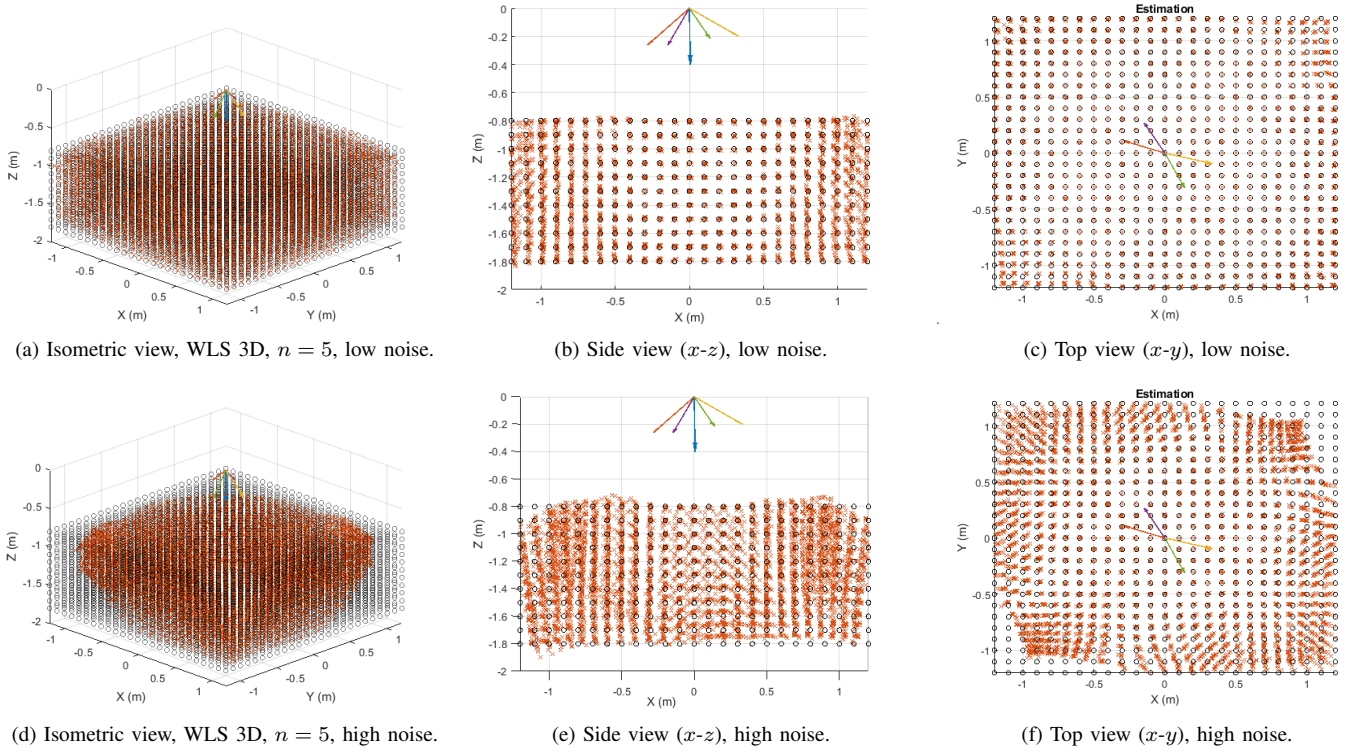


Fig. 7. Reference versus estimated positions in 3-D using WLS with $n = 5$ orientations. Top row: low-noise scenario ($\text{CDF}_{90} = 0.041$ m). Bottom row: high-noise scenario ($\text{CDF}_{90} = 0.21$ m). Columns left to right: (a),(d) isometric perspective; (b),(e) side (y - z) projection; (c),(f) top (x - y) projection.

Fig. 8. GA convergence of CDF_{90} (3D, $n = 6$).

Fig. 9. Optimised orientations ($n = 6$) plotted on the unit sphere around the LED.

coinciding with (17). Multiplying by $C^2 = (\mathbf{n}_t^{(1)} \cdot \mathbf{d})^2$ yields the residual covariance $\Sigma_r = C^2 \Sigma_\beta$ used in GLS. Higher-order terms omitted here are all $O(\sigma^4/N^2)$.

B. Derivation of the Fisher Information Matrix

Let $f(\mathbf{P}_r; \boldsymbol{\theta}) = (2\pi\sigma^2)^{-n/2} \exp[-\|\mathbf{P}_r - \boldsymbol{\mu}(\boldsymbol{\theta})\|^2 / (2\sigma^2)]$ be the likelihood of (??). The log-likelihood is

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2\sigma^2} \|\mathbf{P}_r - \boldsymbol{\mu}(\boldsymbol{\theta})\|^2 + \text{const.}$$

Its gradient is $\nabla_{\boldsymbol{\theta}} \ell = (1/\sigma^2) \sum_{i=1}^n [P_{r,i} - \mu_i(\boldsymbol{\theta})] \nabla_{\boldsymbol{\theta}} \mu_i$. Taking expectation with respect to $\mathbf{P}_r$ and using $\mathbb{E}[P_{r,i}] = \mu_i$ yields $\mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell] = \mathbf{0}$ and

$$\mathbf{I}(\boldsymbol{\theta}) = \mathbb{E}[(\nabla_{\boldsymbol{\theta}} \ell)(\nabla_{\boldsymbol{\theta}} \ell)^T] = \frac{1}{\sigma^2} \sum_{i=1}^n \nabla_{\boldsymbol{\theta}} \mu_i \nabla_{\boldsymbol{\theta}} \mu_i^T,$$

which is (??). The gradient $\nabla_{\boldsymbol{\theta}} \mu_i$ is obtained via the chain rule: $\nabla_{\mathbf{d}} \mu_i$ from (??) followed by $\partial \mathbf{d} / \partial \boldsymbol{\theta} = \mathbf{I}_3$. All steps are first-order differentiations and hence analytically tractable.

REFERENCES

- [1] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory*. Englewood Cliffs, NJ: Prentice Hall, 1993.

Fig. 10. Median angular error (degrees) when only direction is estimated (unknown C).

Fig. 11. Coverage angle vs. CDF_{90} for random vs. GA-optimised orientation sets.